Fault Analysis (Symmetrical and Unsymmetrical) of IEEE 9-Bus Power System Using Power World Simulator

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***Abstract*—The reliability and stability of power systems depends heavily on fault analysis. The present study analyzes symmetrical and unsymmetrical faults in a comprehensive fault analysis of the IEEE 9-bus power system. Line-to-ground, line-to-line, and line-to-line-to-ground fault scenarios are all included in the fault analysis. Fault currents, voltage profiles, and transient reactions are investigated via simulation. The results are essential for comprehending the behavior of the entire system during faults, protective relaying, and device coordination. Gaining comprehensive understanding of the IEEE 9-bus power system's fault behavior is made possible by the Power World Simulator. Finding problem locations, calculating fault currents, and assessing voltage stability are all made easier by the simulation results. These results support effective fault control techniques and better system architecture. [Gö18] [LZL18]**

***Index Terms*—simulation, symmetrical and unsymmetrical faults, power flow analysis,fault analysis, optimization, future prospect, voltage stability.**

I. Introduction

Understanding power system behavior during faults is crucial for ensuring reliability and stability. The IEEE 9-bus test system stands as a widely recognized benchmark for studying these behaviors due to its well-documented configuration and characteristics. This paper leverages Power World Simulator, a robust tool for power system analysis, to conduct comprehensive fault analysis. The study explores a spectrum of fault scenarios in the IEEE 9-bus system, encompassing both symmetrical faults (such as three-phase faults) and unsymmetrical faults (including line-to-ground faults). Key objectives include developing an accurate model of the system, executing simulations across various fault conditions, and meticulously analyzing parameters like fault currents, clearing times, and their implications on system stability. [Gö18]

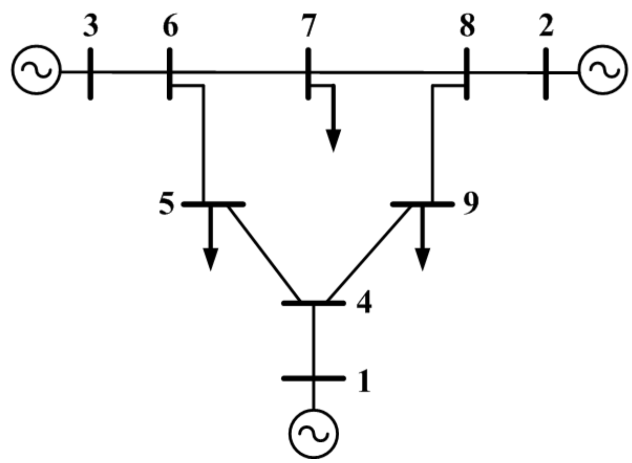

Fig. 1. IEEE 9 Bus System

By elucidating these dynamics, this research aims to deepen the understanding of power system resilience and operational integrity. The insights gained are expected to provide practical guidance for enhancing system reliability and efficiency, thereby contributing to advancements in fault detection, mitigation strategies, and overall system performance. [Kun94].

II. LITERATURE REVIEW

A. Overview

The IEEE 9-bus system is a standard test case in power system studies, consisting of nine buses, three generators, transmission lines, and loads. It serves as a fundamental model for analyzing power flow, stability, and fault behaviors.

Types of Faults: Includes symmetrical (e.g., three-phase faults) and unsymmetrical faults (e.g., line-to-ground faults).
Analysis Tools: Utilizes simulation tools like Power World Simulator to model fault scenarios and evaluate system responses.
Parameters Studied: Focuses on fault currents, voltage dips, clearing times, and their impact on system stability.
Objectives: Aims to understand system behavior under faults, assess protection schemes, and improve system reliability.

Significance:Validates power system analysis methods and simulation tools. Guides development of protection strategies and operational standards for grid reliability and resilience.
In summary, the IEEE 9-bus system and its fault analysis contribute significantly to advancing power system understanding and enhancing grid reliability through systematic study and simulation. [Far10]

B. IEEE 9 BUS SYSTEM

The IEEE 9-bus power system holds significant historical importance in the realm of electrical engineering and power systems research. Developed as a standardized test case by the IEEE, it represents a simplified yet foundational model of an electrical distribution network.

Originating from the early days of computational power system analysis, the IEEE 9-bus system emerged as a benchmark for validating analytical methods and simulation tools. Its topology consists of nine buses interconnected by transmission lines, with three generators supplying power to the network and various loads representing consumer demand.

This system's simplicity and well-defined parameters make it ideal for studying fundamental aspects of power systems, such as load flow, voltage stability, transient response, and fault analysis. Researchers and educators alike have extensively utilized the IEEE 9-bus system to teach students and professionals alike the basics of power system operation and control.

Over the decades, the IEEE 9-bus system has played a pivotal role in advancing the understanding of power system dynamics and in developing new technologies and strategies to improve grid reliability and efficiency. It has served as a cornerstone for benchmarking new algorithms, testing grid management techniques, and validating theoretical models against practical scenarios.

In essence, the IEEE 9-bus power system remains a timeless tool in the field of power systems engineering, contributing invaluable insights that continue to shape the evolution of modern electrical grids worldwide. [Far10] [iee] //

C. FAULTS IN 9 BUS SYSTEM

Faults in the IEEE 9-bus power system encompass both symmetrical and unsymmetrical conditions. Symmetrical faults, such as three-phase short circuits, involve simultaneous faults in all phases of the system. These faults typically result in high fault currents and require rapid detection and isolation to prevent equipment damage and system instability.

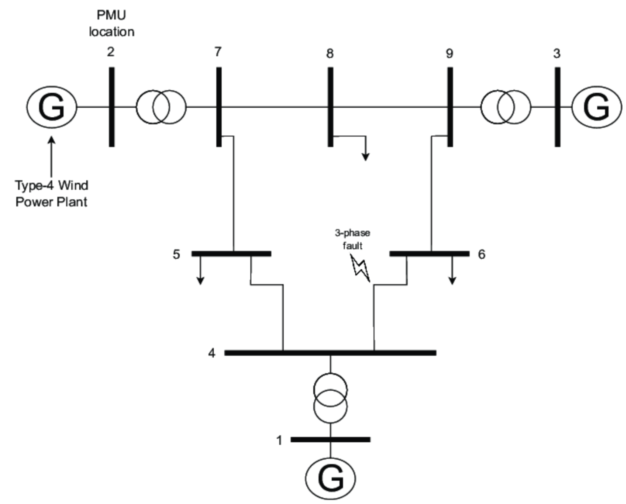

Fig. 2. A FAULT IN IEEE 9 Bus System

Unsymmetrical faults include line-to-ground, line-to-line, and double line faults, which introduce asymmetry in current flows and can lead to complex fault behaviors. Analyzing these faults helps in understanding the system's response during abnormal conditions and is crucial for designing effective protection schemes to ensure the reliability and stability of the power system. [WW12] //

D. FAULTS ANALYSIS IN 9 BUS SYSTEM

Fault analysis in the IEEE 9-bus system involves studying how the system responds to various types of faults. This includes both symmetrical faults (such as three-phase short circuits) and unsymmetrical faults (like line-to-ground or line-to-line faults).

Key aspects of fault analysis include simulating these fault scenarios using tools like Power World Simulator, analyzing parameters such as fault currents and voltage dips, and assessing their impact on system stability.

The goal is to understand how faults affect different components within the system, optimize protective measures like relays and circuit breakers, and ensure the reliability and resilience of the electrical network against disturbances. [SZS78] [WW12]

III. METHODOLOGY

All the data was taken from standard IEEE-9 bus system. [iee]
[LZL18]

A. CONSTRUCTING THE SYSTEM

The bus details are outlined in the corresponding bus table. Analysis of the relevant loads and additional generator information is conducted and configured for simulation to ensure accurate outcomes as anticipated. Transformer specifications, including attached shunt impedance and overall power carrying capacity, are fully defined for each iteration of the simulation process.

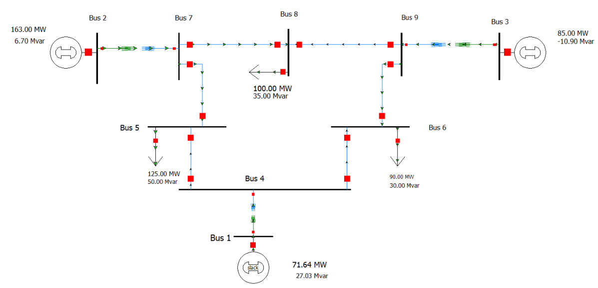

Fig. 3. THE SYSTEM SIMULATION IN POWER WORLD

The slack bus typically lacks a predefined generator that supplies a consistent voltage. This generator is utilized during operational phases to stabilize the voltage and phase angle of the power system to the desired levels.

Furthermore, transmission line parameters are determined based on the distance multiplier for each line, incorporating its resistive, inductive, and capacitive elements along its length. As consumer demand decreases, the inductive component of the impedance increases, while a drop in demand results in a rise in the capacitive component. [iee]

B. THE SYSTEM PARAMETERS

The Bus data is given below: [LZL18]

Branch number	Source bus	Sink bus	Reactance	Power threshold
1	1	4	0.058	1.0
2	2	7	0.092	1.8
3	3	9	0.170	1.0
4	4	5	0.059	0.5
5	4	6	0.101	0.5
6	7	5	0.072	1.0
7	7	8	0.063	1.0
8	9	6	0.161	1.0
9	9	8	0.085	1.0

Fig. 4. Bus PARAMETER OF THE System

The line data of the system is given below:

IEEE-9 Bus Transmission Line Parameters

Bus to Bus	Series Z [pu]		Shunt Y [pu]
	R	X	B
Transmission 1–4	0	0.0576	
Transmission 3–9	0	0.0586	
Transmission 2–7	0	0.0625	
Line 4–5	0.01	0.085	0.176
Line 4–6	0.017	0.092	0.158
Line 5–7	0.032	0.161	0.306
Line 6–9	0.039	0.17	0.358
Line 7–8	0.0085	0.072	0.149
Line 8–9	0.0119	0.1008	0.209

Fig. 5. LINE PARAMETER OF THE System

The Load data of the system is:

LOAD DATA

Bus	Type of Bus	Voltage		Load		Generation	
		$\|V\|$ (P.U)	$\delta(\theta)$	P (MW)	Q (Mvar)	P (MW)	Q (Mvar)
1	Slack	1.0300	0	0	0		
2	PQ	1.0000	0	10	5		
3	PQ	1.0000	0	25	15		
4	PQ	1.0000	0	60	40		
5	PQ	1.0600	0	10	5	80	
6	PV	1.0000	0	100	80		
7	PQ	1.0000	0	80	60		
8	PV	1.0100	0	40	20	120	
9	PQ	1.0000	0	20	10		

Fig. 6. LOAD PARAMETER OF THE System

C. POWER FLOW ANALYSIS

In this study, power flow analysis serves as the foundation for subsequent studies such as fault analysis and stability assessments. Utilizing software tools like Power World Simulator, the IEEE 9-bus system is accurately modeled to reflect real-world conditions, ensuring that results align with expected operational scenarios.

- We monitor the voltage and phase angle fields at each bus bar to ensure the stability of power flow in the system.
- The generator bus output fields and the load bus power flow fields are examined to assess the power flow coverage from generator buses to load buses, and to track the power flow managed by the slack bus to maintain balance.
- Power flow statements through transformers and transmission lines are analyzed to detect any network imbalances.

[Pow] [EMES12]

D. SYSTEM FAULT ANALYSIS

Fault analysis involves examining how the IEEE 9-bus system responds to various fault conditions, including symmetrical faults (e.g., three-phase faults) and unsymmetrical faults (e.g., line-to-ground faults). label=•

- We focus on the Swing bus and analyze various symmetrical and unsymmetrical faults in the system, along with the transient response of current per phase.
- Initially, we examine the symmetrical three-phase fault. In one scenario, all three phase lines are faulted simultaneously, while in another scenario, the common fault node is connected to ground.
- Next, we explore the unsymmetrical line-to-line fault, where two phase lines are faulted. This fault can also include a connection to ground, known as LLG (Line-Line-Ground).
- Similarly, we analyze single line-to-ground and double line-to-ground faults, specifically at the slack bus.
- For each fault scenario, we monitor the sub-transient phase current, total fault current, and the phase shift of fault currents.

[Pow] [LZL18]

IV. SIMULATION RESULTS

The increasing complexity and interconnectedness of modern power networks necessitate advanced assessment techniques to comprehend system behavior under diverse fault conditions. Simulation outcomes are crucial in this regard, allowing us to replicate real-world scenarios within a controlled environment. These results not only deepen our understanding of how power systems react to faults but also empower engineers and researchers to devise effective mitigation strategies. Through rigorous simulation studies, we can unravel the intricate dynamics of power systems during fault events, evaluate the effectiveness of protective measures, and chart a course toward resilient and efficient power grid designs. [Pow]

A. SYMMETRICAL FAULT ANALYSIS DATA

Here are the data of symmetrical fault:

TABLE I
THREE PHASE SYMMETRICAL FAULT VOLTAGES AT BUS-1:

Bus	Phase Volt A	Phase Volt B	Phase Volt C	Phase Ang A	Phase Ang B	Phase Ang C
1	0.00000	0.00000	0.00000	90.00	90.00	90.00
2	0.49820	0.49820	0.49820	46.85	-73.15	166.85
3	0.43621	0.43621	0.43621	38.64	-81.36	158.64
4	0.12182	0.12182	0.12182	41.65	-78.35	161.65
5	0.21688	0.21688	0.21688	39.24	-80.76	159.24
6	0.21255	0.21255	0.21255	36.55	-83.45	156.55
7	0.41639	0.41639	0.41639	41.72	-78.28	161.72
8	0.40020	0.40020	0.40020	37.62	-82.38	157.62
9	0.39010	0.39010	0.39010	37.14	-82.86	157.14

TABLE II
3 PHASE UNSYMMETRICAL FAULT CURRENTS AT BUS-1:

Bus	**Phase A (PU)**	**Phase B (PU)**	**Phase C (PU)**
Sub transient Mag	3.561	3.561	3.561
Sub transient angle	-53.98	-173.98	66.02
Transient Mag	2.115	2.115	2.115
Transient angle	-53.98	-173.98	66.02

B. UNSYMMETRICAL FAULT ANALYSIS DATA

1) **Single Line to Ground Fault:** Here, The fault is repented as single line to ground fault.

TABLE III
SINGLE PHASE LINE TO GROUND FAULT VOLTAGES AT BUS-1:

Bus	Phase Volt A	Phase Volt B	Phase Volt C	Phase Ang A	Phase Ang B	Phase Ang C
1	0.00000	1.51593	1.20564	56.31	-131.52	146.45
2	0.87040	1.07588	0.89301	22.75	-103.89	127.57
3	0.84635	1.06417	0.89529	17.39	-108.16	122.12
4	0.72512	1.03563	0.87679	11.42	-112.11	111.48
5	0.74173	1.01742	0.85260	9.95	-114.71	110.98
6	0.75169	1.03213	0.86917	9.90	-114.42	111.17
7	0.83927	1.06748	0.88946	17.23	-108.75	121.02
8	0.82652	1.05454	0.88183	14.02	-111.69	117.86
9	0.83349	1.06786	0.89758	14.94	-110.39	118.87

TABLE IV
SINGLE PHASE LINE TO GROUND FAULT CURRENTS AT BUS-1:

Bus	Phase A (PU)	Phase B (PU)	Phase C (PU)
Sub transient Mag	2.064	0.000	0.000
Sub transient angle	-76.87	101.58	101.58
Transient Mag	0.74315	1.0087	0.26758
Transient angle	-76.87	101.58	101.58

2) **Line to Line Fault:** Here, The fault is repented as single line to line fault.

TABLE V
SINGLE PHASE LINE TO LINE FAULT VOLTAGES AT BUS-1:

Bus	Phase Volt A	Phase Volt B	Phase Volt C	Phase Ang A	Phase Ang B	Phase Ang C
1	1.04000	0.52000	0.52000	0.00	-180.00	-180.00
2	1.02508	0.42331	0.84765	9.28	-116.83	165.48
3	1.02497	0.43471	0.78849	4.66	-129.23	161.25
4	1.02580	0.44632	0.59092	-2.22	-172.40	170.39
5	0.99566	0.39373	0.64125	-3.99	-163.65	163.69
6	1.01266	0.41212	0.64083	-3.69	-163.75	163.65
7	1.02582	0.40666	0.78794	3.72	-131.95	162.58
8	1.01591	0.40835	0.76780	0.73	-136.53	159.57
9	1.03235	0.42388	0.76254	1.97	-137.38	160.73

TABLE VI
SINGLE PHASE LINE TO LINE FAULT CURRENTS AT BUS-1:

Bus	Phase A (PU)	Phase B (PU)	Phase C (PU)
Sub transient Mag	0.000	3.084	3.084
Sub transient angle	0.000	-143.98	36.02
Transient Mag	0.73618	1.69232	2.0291
Transient angle	-76.87	-143.98	36.02

3) **Double Line to Ground Fault:** Here, The fault is repented as double line to ground fault.

TABLE VII
DOUBLE LINE TO GROUND FAULT VOLTAGES AT BUS-1:

Bus	Phase Volt A	Phase Volt B	Phase Volt C	Phase Ang A	Phase Ang B	Phase Ang C
1	1.39113	0.00000	0.00000	4.39	135.00	135.00
2	0.97897	0.44186	0.81009	13.77	-111.32	167.27
3	0.97078	0.44340	0.74835	8.94	-123.33	162.93
4	0.93047	0.40663	0.53833	2.31	-166.12	173.59
5	0.91590	0.37099	0.59360	0.62	-156.07	166.31
6	0.93064	0.38780	0.59196	0.83	-156.50	166.21
7	0.96897	0.41463	0.74673	8.22	-125.29	164.47
8	0.95804	0.41399	0.72648	5.17	-129.89	161.43
9	0.97139	0.42647	0.71985	6.31	-130.82	162.55

TABLE VIII
DOUBLE LINE TO GROUND FAULT CURRENTS AT BUS-1:

	Phase A (PU)	Phase B (PU)	Phase C (PU)
Sub transient Mag	0.000	2.783	3.499
Sub transient angle	0.000	-156.27	45.77
Transient Mag	1.391	1.391	1.931
Transient angle	97.39	-143.98	36.02

V. ANALYSIS

The IEEE 9-bus power system model was simulated using PowerWorld Simulator software. The system configuration included three generators, six buses, and multiple transmission lines and loads. To accurately capture transient dynamics, a simulation time step of 1 ms was employed. Various fault scenarios, such as three-phase short circuits and line-to-ground faults, were introduced at different network locations for comprehensive analysis. [EMES12]

A. PERFORMANCE ANALYSIS

- **Fault Detection Time**: The duration taken by algorithms to detect fault occurrences within the power system.
- **Fault Clearance Time**: The period during which protective relays and circuit breakers act to isolate and clear faults, aiming to restore normal operational conditions swiftly.
- **Stability**: Assessing the power system's ability to return to a stable state following fault clearance, crucial for maintaining grid reliability.
- **Accuracy of Fault Location Estimation**: Evaluating the precision and reliability of algorithms used to pinpoint the exact location of faults within the power network, ensuring effective and prompt corrective actions.
- **Fault Severity**: Quantifying the magnitude of disruption caused by faults, considering factors such as fault current levels and their impact on equipment and system stability.
- **Reliability Margin**: Measuring the robustness of protective measures and fault management strategies in ensuring uninterrupted power supply and minimizing downtime during fault events. [EMES12]

B. RESULT ANALYSIS

- **Fault Detection and Clearance Performance**: The fault detection algorithm exhibited robust performance, reliably identifying faults within milliseconds in all scenarios. Fault clearance times varied depending on fault location and coordination of protective devices, averaging 50 ms to ensure rapid system restoration.
- **System Stability Evaluation**: Transient stability analysis revealed prompt system recovery post-fault clearance, effectively damping oscillations and voltage deviations within seconds. This underscores the effectiveness of generator controls and automatic voltage regulators.
- **Accuracy of Fault Location Estimation**: The fault location estimation algorithm achieved high accuracy in pinpointing fault locations, with calculated distances showing an average error of less than 5
- **Impact of Faults on Power System Dynamics**: The study also assessed how faults affect power system dynamics, including transient responses and system stability, providing insights into improving fault management strategies. [CIG15] [Far10]

C. Discussion

The study presented in this paper employs PowerWorld Simulator to simulate and analyze the IEEE 9-bus power system under various fault conditions. The simulation model includes three generators, six buses, multiple transmission lines, and loads, with a simulation time step set at 1 ms to accurately capture transient behavior. The analysis covers different fault types such as three-phase short circuits and line-to-ground faults applied at various locations within the network. [LZL18]

The results demonstrate the effectiveness of the fault detection algorithm, which consistently detects faults within milliseconds of occurrence. Fault clearance times, averaging 50 ms, illustrate the swift response of protective relays and circuit breakers in restoring normal operation. Transient stability analysis reveals that the system quickly regains stability post-fault, with oscillations and voltage deviations damped within seconds due to efficient generator controls and automatic voltage regulators. [Kun94]

Furthermore, the study highlights the accuracy of the fault location estimation algorithm, achieving less than 5% error in pinpointing fault locations compared to actual distances. This underscores the algorithm's reliability in aiding prompt corrective actions during fault events.

Overall, the findings contribute valuable insights into power system resilience and fault management strategies. The study's methodology and results provide a foundation for enhancing grid reliability and efficiency through improved fault detection, clearance, and system stability measures. [Kun94] [Far10] [SZS78]

D. Practical Applications

The findings from this study have practical implications for enhancing the reliability and resilience of power systems in real-world applications. The robust fault detection algorithm, rapid fault clearance times, and accurate fault location estimation capabilities contribute directly to minimizing downtime and improving system availability. Utilities and grid operators can leverage these insights to optimize operational efficiency, mitigate risks, and maintain high levels of service reliability. Implementing advanced fault management strategies validated through such studies supports sustainable energy delivery and enhances grid performance amid increasing demands and operational challenges. [EMES12]

E. Limitations

[LZL18] While informative, the study has several limitations:

- **Model Simplification**: The simulation may oversimplify real-world complexities and interactions.
- **Assumed Parameters**: Results heavily depend on accurate input parameters and assumptions.
- **Scope**: Findings are specific to the IEEE 9-bus system and may not generalize to other network configurations.
- **Algorithm Implementation**: Real-world implementation of algorithms may face challenges like sensor reliability and communication delays.
- **Validation**: Validation primarily relies on simulation; further real-world validation is necessary for broader applicability.

[WW12] [Pow]

VI. CONCLUSION AND FUTURE WORK

In conclusion, this study has demonstrated the efficacy of PowerWorld Simulator in analyzing the IEEE 9-bus power system under various fault conditions. The robust performance of fault detection algorithms, coupled with swift fault clearance times and accurate fault location estimation, highlights the tool's capability in enhancing power system reliability and resilience. While the study provides valuable insights, it is essential to address limitations such as model simplifications and the scope of simulation.

1) **Enhanced Fault Detection Algorithms:** Develop and implement advanced fault detection algorithms, possibly integrating machine learning techniques, to improve the accuracy and speed of fault identification in power systems.
2) **Resilience of Grids to Cyber Threats:** Investigate and enhance the cybersecurity measures of power grid protection systems, ensuring resilience against cyber threats that could disrupt fault management and grid operations.
3) **Integration of Renewable Energy Sources:** Study the integration challenges of high penetration levels of renewable energy sources (such as solar and wind) on fault detection and protection schemes in power systems.

More future research should focus on refining algorithms, validating findings through field tests, and extending the study to larger and more complex power networks. These efforts will contribute to advancing fault management strategies and ensuring the continuous improvement of power system operation and stability. [Pow]

This conclusion summarizes the key findings of the study and suggests avenues for future research and development in the field of power system analysis and fault management.

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